

Name \_\_\_\_\_

## Quiz 4

Fill in the missing parts of the proof by induction of the theorem below.

**Theorem:** For every finite undirected graph, the sum of all vertex degrees is equal to twice the number of edges.

**Proof:** (2 points) Universal declaration: Let  $G$  be an arbitrary

We induct on  $m$ , the number of edges in  $G$ .

(3 points) Induction hypothesis:

There are two cases:

- Base case: Suppose  $m = 0$ . Then every degree is zero, so the sum of the degrees is  $0 = 2 \cdot 0$ , so the claim holds.
- (5 points) Inductive case: Suppose

The two cases are exhaustive, and the claim holds in both cases. Therefore, the claim that every finite undirected graph with  $m$  edges has a sum of vertex degrees equal to  $2m$  holds for every nonnegative integer  $m$ .

□